

Automatic PLANTS WATERING SYSTEM With Melody

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Here is a circuit for automatic plants watering, which can be undertaken every morning without any human effort. A sensor is used to detect ambient light and activate a pump motor to start watering the plants in the morning. You can set the watering time duration as per your requirement. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the automatic plants watering system with melody is shown in Fig. 2. It is built around NE555 timer (IC1 and IC3), source driver UDN2981 (IC2), melody generator UM66 (IC4), 9V voltage regulator 7809 (IC5),



Fig. 1: Author's prototype

nnp transistor BC547 (T1), light-dependent resistor (LDR1) and a few other components.

NE555 (IC1) is configured as a monostable multivibrator. It acts as a one-shot pulse generator. Output pulse is available at its output pin 3 of IC1. Pulse begins when IC1 receives a low trigger input at pin 2, which falls below 1/3rd of its supply voltage.

Output pulse ends when voltage on capacitor C1 equals 2/3rd supply voltage. That is, width of output pulse depends on the time constant of resistor R2 and capacitor C1. This is determined by the following relationship:

$$T = 1.1 \times R2 \times C1$$

UDN2981 (IC2) is the source

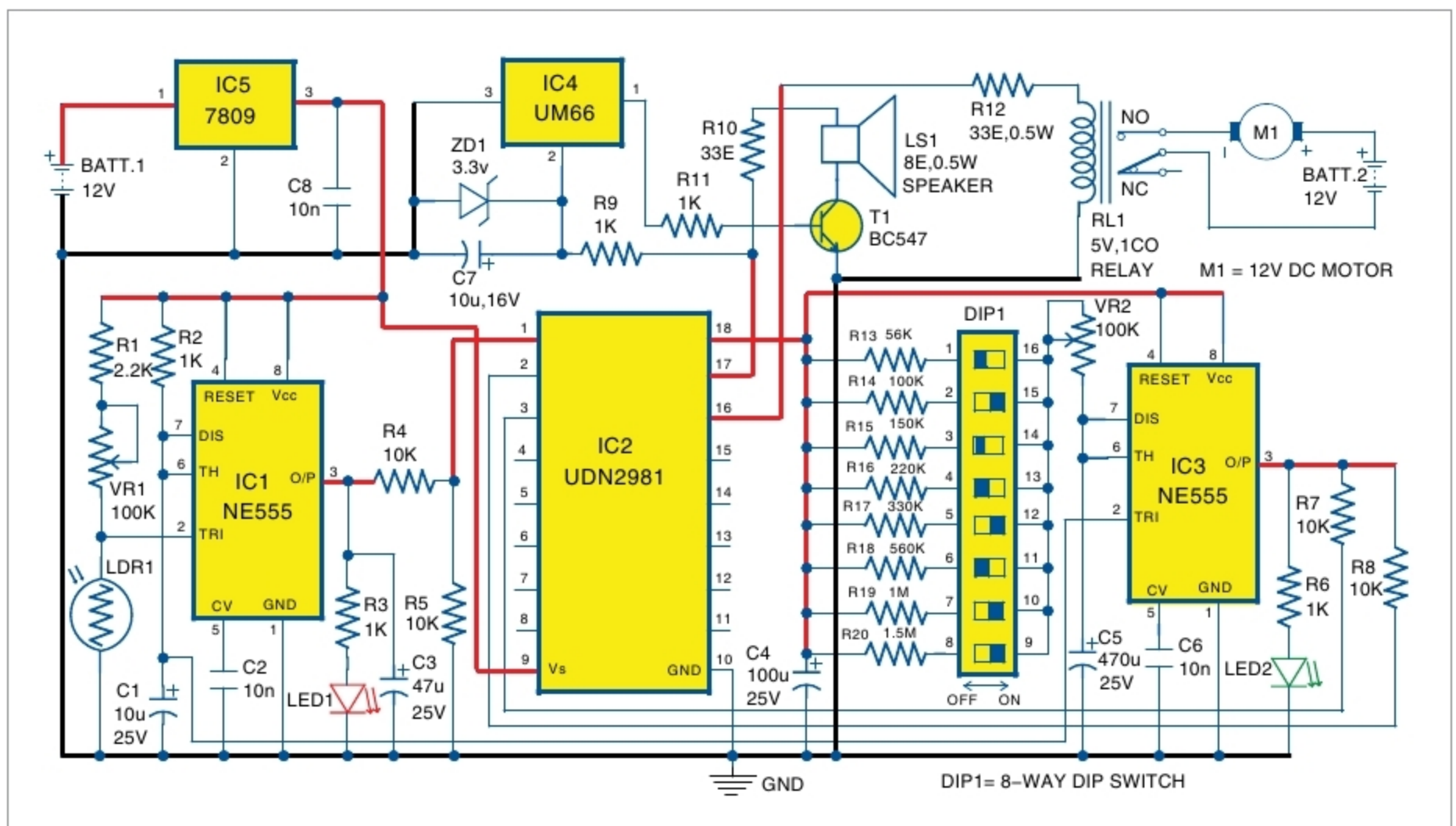


Fig. 2: Circuit diagram of automatic plants watering system with melody